ASSIGNMENT No. 07

Searching and Sorting Algorithms: *Implement Linear Search, Binary Search, Bubble Sort, Selection Sort, and Insertion Sort. Compare their performance using different input sizes and analyze their time complexity.*

1. Linear Search and Binary Search.
2. Bubble Sort.
3. Selection Sort.
4. Insertion Sort.
5. A student owns three versions of a book, containing A, B, and C pages, respectively. One day, the student locked his/her room and went to the market. Upon returning, he/she realized that he/she had forgotten the password X to open the room. Fortunately, the student remembered a clue: a number N. The password X is defined as follows:
   • Consider all positive numbers that are divisible by A, B, or C.
   • Arrange these numbers in their natural order (from smallest to largest) as they appear in the sequence without any duplicate numbers.
   • The password X is the number that appears at the N-th position in this sequence.
   Write a C program to find the password X, given the values of A, B, C, and N.
   Example:
   Input: A = 3, B = 5, C = 7, N = 10
   Step 1: List numbers divisible by A, B, or C in increasing order: 3, 5, 6, 7, 9, 10, 12, 14, 15, 18, . .
   Step 2: Find the N-th element: The 10th element is 18 Output: X = 18